Curriculum Vitae

Remi Seddigh

PhD Candidate in Theoretical Physics

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Center for Quantum Mathematics and Physics (QMAP) · University of California, Davis, CA 95616

Education

PhD in Physics 2023 to 2028 (expected)
University of California, Davis

Advised by Prof. Veronika Hubeny at the Center for Quantum Mathematics and Physics (QMAP). Qualifying examination passed in November 2025.

MA in Physics 2024
University of California, Davis

BA in Physics, minor in Computer Science 2023
University of California, Berkeley

Fellowships and Awards

Department of Physics graduate fellowship, part-time Winter 2025, Fall 2026
University of California, Davis

Distinction in General Scholarship 2023
University of California, Berkeley

Research Experience

Graduate Student Researcher, FSG Group 2024 to present
QMAP, University of California, Davis

Research as part of the Fields, Strings, Gravity group, with Prof. Veronika Hubeny, on the holographic entropy cone, the set of entanglement entropy patterns that holographic states can realize, using graph models. In a graph model, the entropy of a boundary subsystem is the cost of the minimum cut homologous to it. Comparing the costs of competing cuts yields linear inequalities in the edge weights, which partition weight space into cells (W-cells) on which the minimal cuts are fixed. We are studying the geometry, combinatorics and topology of the resulting cell complexes and their relation to the lattice of mutual information patterns of the subsystems (the KC lattice).

Research Assistant, Jacak Group Summer 2022
Lawrence Berkeley National Laboratory

Worked on the ATHENA detector for the Electron-Ion Collider. Designed and tested cooling local support staves, evaluating carbon foams across temperatures and airflows to establish the feasibility of air cooling through carbon foam.

Research Apprentice, Pyle Group 2020 to 2022
University of California, Berkeley

Worked on the SuperCDMS low-mass dark matter search with Prof. Matt Pyle and Dr. Bruno Serfass: design optimization of massive low-temperature calorimeters, component design, and data analysis. Built the group's data repository and website in Python, SQL and Pyramid.

Papers in Preparation

W-Complexes, S-Complexes, and the KC Lattice
(with Veronika Hubeny and Massimiliano Rota)

Talks and Presentations

Min-Cut Structures of Holographic Graph Models April 2026
FSG Group Journal Club, UC Davis

Entanglement Wedge Cross-Sections — March 2026
Physics 233 (Black Holes and AdS/CFT) final presentation, UC Davis

The Strong Cosmic Censorship Conjecture — June 2025
Physics 233 (Advanced General Relativity) final presentation, UC Davis

Graph Models and W-Cells — February 2025
FSG Group Journal Club, UC Davis

Entanglement Entropy Evolution During Gravitational Collapse — February 2025
FSG Group Journal Club, UC Davis

The Holographic Entropy Cone — January 2025
QMAP Journal Club, UC Davis

RT and HRT Surfaces — September 2024
QMAP Journal Club, UC Davis

Conferences Attended

Future of Mathematics Symposium — May 2026
Stanford University

BASICS 2025, Bay Area Strings, Information and Cosmology Symposium — November 2025
University of California, Davis (co-organizer)

BASICS 2024 — November 2024
University of California, Berkeley

Teaching

Associate Instructor, Physics 7C — Summer Session I 2026
University of California, Davis

Waves, optics, quantum mechanics and electromagnetism, for 161 students across six discussion lab sections, with five teaching assistants. Wrote and delivered the lectures and demos, wrote and graded both midterms, and created the practice problems. Student evaluations were 4.6 of 5 for overall teaching effectiveness and 4.7 for learning environment and enthusiasm.

Teaching Assistant, Department of Physics — 2023 to present
University of California, Davis

Teaching assistant for Physics 7ABC (physics for biology and medical majors), 9A (mechanics for engineering majors), 9HD (electricity and magnetism for physics majors), 10 (climate science), 105A (upper-division mechanics) and 200A (graduate classical mechanics), as discussion, lab and course TA.

Instructor, Peninsula Bridge — Summer 2018
Belmont, California

Taught science, music and computer programming to seventh-grade students in a summer program for underserved students.

Service and Outreach

Co-organizer, BASICS 2025 — November 2025

Bay Area Strings, Information and Cosmology Symposium, UC Davis.

Executive Board Member for Cultural Activities — 2020 to 2023

ISCO Students' Cultural Organization, UC Berkeley.

Skills

Programming Python (NumPy, SciPy, Matplotlib, Monte Carlo simulation, Jupyter), Normaliz, Mathematica, C, Java, SQL. LLM use for mathematics and software development. Familiar with ROOT, Scheme and RISC-V assembly, as well as LLM architecture. UNIX on macOS and Linux.

Methods Geometric combinatorics, differential geometry, graph theory, complex analysis, probability and statistics. Experimental laboratory work and LabVIEW.

Languages English, French and Persian, all fluent. German at intermediate level, some Mandarin.